Nathaniel Johnston

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Work Experience

PCB Designer **Jun. 2025 - Pres.**
Apple

- Used **Cadence Allegro** to complete layout of high speed and high power PCBs from 4 to 26 layers
- Completed layout of flex PCBs with up to 4 layers with high power and signal integrity requirements
- Routed **SOC** boards with high speed signals such as **PCIe 6.0** and **USB4**, and data rates of over **60 Gbps**
- Worked with PCB fabricators to resolve design issues throughout the design process
- Wrote custom scripts using the SKILL programming language to create custom features and increase work efficiency

PCB Layout Designer & Components Librarian **Jan. 2023 - May 2025**
Kepler Communications

- Used **Altium** to design, layout, and simulate a power delivery board capable of over **300W** for space applications
- Applied **DFM** and **DFT** principles to complete layout of complex 4-8 layer PCBs, often with high speed or high power
- Fully tested and qualified custom buck-boost power regulation boards
- Made hundreds of schematic symbols and PCB footprints to **IPC** standards
- Sped up designs and reduced workload with custom **C#** Altium extension that performed automated design checks
- Led migration of 7000+ components from SQL DB to Altium 365 and implemented consistent naming convention

Solutions Architecture Developer **May 2022 - Aug. 2022**
BlackBerry Ltd.

- Worked as sole developer for a Syslog emulator to generate all types of BlackBerry Syslogs for test and demo use cases
- Designed a **Python** parser to convert Syslogs into JSON using a combination of RegEx and string methods
- Developed systems using **Docker** so the code could be easily run on any computer
- Wrote multiple blog posts about BlackBerry technologies on BlackBerry's developer blog

Solutions Architecture Developer **Sep. 2021 - Dec. 2021**
BlackBerry Ltd.

- Aided in design of **React.js** application to allow users to automatically find security vulnerabilities in mobile apps
- Added and modified server side **REST** endpoints using **Node.js**
- Used **React Native** and **Firebase** to build an **Android** app to receive security threat notifications on remote devices

Hardware Designer - Research Assistant **Jan. 2021 - Apr. 2021**
Institute for Quantum Computing - University of Waterloo

- Used **KiCad** to complete schematic capture and layout of an RF power amplifier PCB for use in a quantum simulator
- Designed and simulated 3rd order maximally flat (Butterworth) RF filters using **LTspice**
- Sourced components and wrote in depth guides and documentation

Motor Control Subteam Lead **Jul. 2019 - Apr. 2020**
Waterloop Student Design Team

- Managed team of 6 students, taught important concepts, and supervised progress through weekly meetings
- Designed high power motor control board with 3-phase IGBT transistor inverter for linear induction motor
- Performed schematic capture and circuit board layout using **Altium Designer** and **KiCad**

Enterprise Solutions Developer - IoT **Sep. 2019 - Dec. 2019**
BlackBerry Ltd.

- Sole designer and developer for proof of concept for smart security system
  - Microcontroller used ultrasonic sensor and camera to detect motion and send video to **Raspberry Pi** via **MQTT**
  - Used **Node.js** and **Python** on a **Raspberry Pi** to detect faces and send secure alerts to users via BlackBerry APIs
  - System automatically disarmed when it detected a nearby familiar device or correct PIN was entered into keypad
- Wrote multiple technical articles on the BlackBerry developer blog
- Served as Subject Matter Expert on new BlackBerry REST API for both company partners and team members

Education

University of Waterloo **Sep. 2017 - Apr. 2022**
Bachelor of Applied Science, Electrical Engineering *Dean's Honors List*

Technical Skills

Hardware: IPC CID certified, Cadence Allegro, Altium Designer, KiCAD, LTSpice, Soldering, Oscilloscope
Languages: Python, C#, Java, JavaScript, C/C++, SQL, Matlab
Embedded: PCIe, SPI, I2C, UART, JTAG, CAN, ESP32, Arduino, Raspberry Pi, TI Launchpad
Other: Linux, Git, Docker, Node.js, React.js, REST, MQTT, Autodesk Inventor